

AI Resume Analyzer and Job Recommendation System

Short Project Report

1. Project Objective

An NLP-based application that helps students understand how well their resumes match selected job roles. It provides match scores, missing skills, and a learning roadmap.

2. Problem Statement

The system addresses the difficulty students face in understanding whether their resumes contain skills expected for particular roles. It focuses on job-related skills, projects, education, and relevant experience.

3. Project Workflow

Step	Process
1	Upload PDF or DOCX resume
2	Extract resume text
3	Clean and normalize text
4	Identify job-related skills
5	Load job-role requirements
6	Compare resume with job roles
7	Calculate match scores
8	Recommend top roles
9	Show missing skills and learning roadmap

4. Technologies Used

Programming: Python

Resume extraction: pypdf, python-docx

Matching: TF-IDF and cosine similarity using scikit-learn

Data handling: Pandas and NumPy

User interface: Streamlit

Visualization: Plotly

Version control: Git and GitHub

5. Main Modules

- Resume Upload
- Text Extraction and Cleaning
- Skill Extraction
- Job Role Dataset
- Matching and Recommendation
- Skill-Gap Analysis
- Learning Roadmap
- Streamlit Dashboard

6. Features Implemented

PDF/DOCX upload, text extraction and cleaning, technical skill detection, comparison with multiple job roles, match scoring, top-three role display, missing-skill analysis, learning roadmap, Streamlit interface, custom job-description comparison, testing sheet, and architecture workflow diagram.

7. Testing

The testing sheet covers PDF/DOCX uploads, skill detection, role matching, top-three recommendations, missing skills, roadmap generation, custom job-description analysis, unsupported file types, and job descriptions without detectable skills.

8. Responsible AI

The system is intended for guidance rather than automatic hiring or rejection. It focuses on job-related information and does not intentionally evaluate protected personal attributes. Match scores are estimates, and a missing keyword does not necessarily mean the underlying ability is absent.

9. Limitations

Keyword-based skill extraction can miss skills expressed using different terminology. TF-IDF similarity may not fully capture semantic similarity. The role dataset represents only the roles and skills included in the dataset.

10. Conclusion

The AI Resume Analyzer provides an end-to-end educational workflow for resume parsing, skill extraction, job-role matching, skill-gap identification, and learning-roadmap generation through a Streamlit dashboard.